

# Solving problems with quadratic equations

Turning words into a quadratic — and deciding which of the two roots the problem actually allows.

LESSON 72–73

2 × 40 MIN

ALGEBRA 8, CHAPTER III \$25

STAGE 9 · 4.1

LESSON CLOCK

5'

12'

8'

13'

2'

|             |        |
|-------------|--------|
| Warm-up     | 5 min  |
| Explanation | 12 min |
| Interactive | 8 min  |
| Practice    | 13 min |
| Homework    | 2 min  |

LEARNING OBJECTIVES

- Choose the unknown and write the equation.
- Solve and interpret both roots in the context.
- Reject a root that has no meaning in the problem.
- Handle the standard types: area, consecutive numbers, speed and work.

TEXTBOOK

|           |                          |
|-----------|--------------------------|
| National  | Chapter III, pp. 153–158 |
| Cambridge | Learner’s Book pp. 84–90 |
| Workbook  | Workbook 4.1             |

01

## Explanation

### The method

- Name the unknown.** Write “Let  $x$  be ...” and say what unit it is in.
- Express everything else** in terms of  $x$ .
- Write the equation** from the sentence that has not been used yet.
- Solve** it.
- Check both roots against the problem** and reject any that make no sense.
- Answer in words**, with units.

STEP 5 IS THE ONE THAT IS NEW

A quadratic gives two roots, but a length, a speed, a time or a number of people cannot be negative. Say explicitly which root you are rejecting and why — it carries a mark.

### The four standard types

**Area.** A rectangle is 3 m longer than it is wide and has area 40 m<sup>2</sup>. Let the width be  $x$ ; then  $x(x + 3) = 40$ , giving  $x = 5$  or  $x = -8$ . A width cannot be negative, so the

rectangle is  $5\text{ m}$  by  $8\text{ m}$ .

**Consecutive numbers.** Two consecutive whole numbers have product 156. Let them be  $x$  and  $x + 1$ :  $x^2 + x - 156 = 0$ , so  $x = 12$  or  $x = -13$ . Both are valid here — 12 and 13, or  $-13$  and  $-12$ .

**Speed.** Distance is  $s = vt$ , so a change of speed on a fixed distance gives a fractional equation that clears to a quadratic.

**Pythagoras.** A right triangle with one leg  $x$  and the other  $x + 7$  and hypotenuse 13:  $x^2 + (x + 7)^2 = 169$ .

### A worked speed problem

A cyclist covers 60 km. If the speed were 5 km/h greater, the journey would take one hour less. Find the speed.

Let the speed be  $x$  km/h,  $x > 0$ . The times are  $\frac{60}{x}$  and  $\frac{60}{x + 5}$ , and the first is one hour longer:

$$\frac{60}{x} - \frac{60}{x + 5} = 1$$

$$60(x + 5) - 60x = x(x + 5)$$

$$300 = x^2 + 5x \implies x^2 + 5x - 300 = 0$$

$$x = 15 \text{ or } x = -20 \text{ — a speed cannot be negative}$$

The speed is  $15\text{ km/h}$ .

## 02 Worked examples

**EXAMPLE 1** A rectangle is 3 m longer than it is wide and has area 40 m<sup>2</sup>. Find its dimensions.

① Let the width be  $x$  m; the length is  $x + 3$  m.

②  $x(x + 3) = 40$   
Area.

③  $x^2 + 3x - 40 = 0$

④  $x = 5$  or  $x = -8$

⑤ A width cannot be negative, so  $x = 5$ .  
Reject the other root.

ANSWER

5 m by 8 m

**EXAMPLE 2** The product of two consecutive whole numbers is 156. Find them.

① Let them be  $x$  and  $x + 1$ .

②  $x(x + 1) = 156$

③  $x^2 + x - 156 = 0$

④  $x = 12$  or  $x = -13$   
Both give whole numbers here.

ANSWER

12 and 13, or  $-13$  and  $-12$

**EXAMPLE 3** A cyclist covers 60 km. Riding 5 km/h faster would save an hour. Find the speed.

① Let the speed be  $x$  km/h.

② 
$$\frac{60}{x} - \frac{60}{x+5} = 1$$

The time saved.

③ 
$$x^2 + 5x - 300 = 0$$

Multiply by  $x(x+5)$ .

④ 
$$x = 15 \text{ or } x = -20$$

Reject the negative speed.

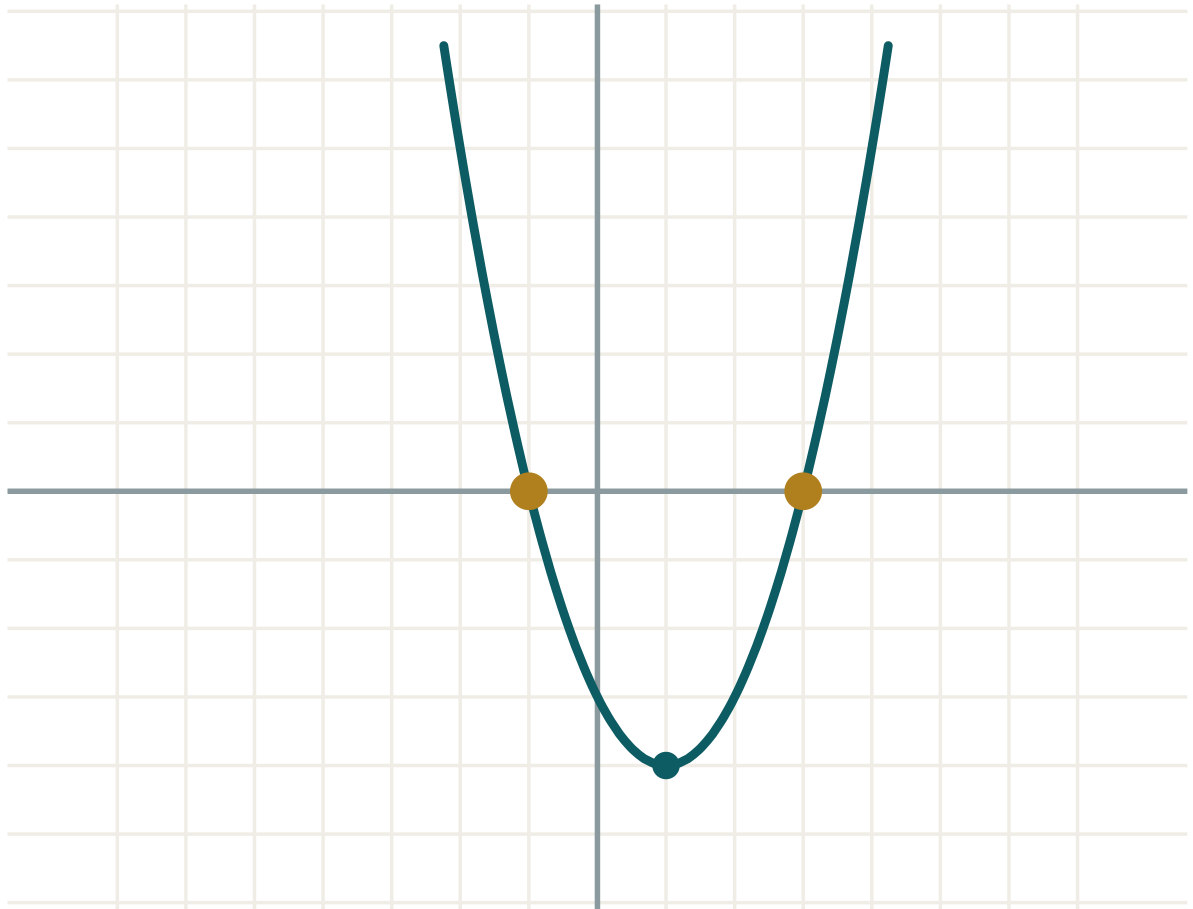
ANSWER

15 km/h

### 03 Interactive model

Write the equation on the board and let the model show which root the graph gives.

## The equation behind the problem



a 1

b -2

c -3

 $D = b^2 - 4ac$  16

roots two

 $x_1$  -1 $x_2$  3 $x_1 + x_2$  2 $x_1 \cdot x_2$  -3

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH              | O'ZBEKCHA           | РУССКИЙ                |
|----------------------|---------------------|------------------------|
| Word problem         | Matnli masala       | Текстовая задача       |
| Let $x$ be...        | $x$ deb belgilaymiz | Пусть $x$ —            |
| Set up an equation   | Tenglama tuzish     | Составить уравнение    |
| Reject a root        | Ildizni rad etish   | Отбросить корень       |
| Consecutive numbers  | Ketma-ket sonlar    | Последовательные числа |
| Average speed        | O'rtacha tezlik     | Средняя скорость       |
| Dimensions           | O'lchamlar          | Размеры                |
| Interpret the answer | Javobni izohlash    | Истолковать ответ      |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

A rectangle has width  $x$  and length  $x + 4$  and area 45. The equation is:

A length comes out as  $x = -6$  or  $x = 4$ . You should:

Two consecutive numbers are written as:

Time equals:

$$\text{distance} \times \text{speed}$$

$$\frac{\text{distance}}{\text{speed}}$$

$$\frac{\text{speed}}{\text{distance}}$$

$$\text{distance} + \text{speed}$$

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS



1. *A square has area  $81 \text{ cm}^2$ . Find its side.*

ANSWER  $9 \text{ cm}$

2. *A number squared is 49. Find it, given it is positive.*

ANSWER  $7$

3. *Two consecutive numbers have product 20. Find them.*

ANSWER  $4$  and  $5$  (or  $-5$  and  $-4$ )

4. *A rectangle is  $2 \text{ m}$  longer than wide, area  $24 \text{ m}^2$ . Write the equation.*

ANSWER  $x(x + 2) = 24$

5. *Solve that equation.*

ANSWER  $x = 4$ , so  $4 \text{ m} \times 6 \text{ m}$

6. *A number added to its square gives 12. Find the positive number.*

ANSWER  $3$

7. *The area of a square is 144. Find the perimeter.*

ANSWER  $48$

**Medium · the standard question**

7 PROBLEMS

1. *A rectangle is 3 m longer than wide and has area 40 m<sup>2</sup>. Find its dimensions.*

ANSWER *5 m × 8 m*

2. *The product of two consecutive whole numbers is 156. Find them.*

ANSWER *12 and 13*

3. *A right triangle has legs  $x$  and  $x + 7$  and hypotenuse 13. Find the legs.*

ANSWER *5 and 12*

4. *A number is 3 less than its square. Find both values.*

ANSWER  $x^2 - x - 3 = 0$ :  $x = \frac{1 \pm \sqrt{13}}{2}$

5. *The sum of a number and its reciprocal is 2.5. Find the number.*

ANSWER  $x = 2$  or  $x = 0.5$

6. *A rectangle has perimeter 26 cm and area 40 cm<sup>2</sup>. Find its sides.*

ANSWER *5 cm and 8 cm*

7. *Two consecutive even numbers have product 168. Find them.*

ANSWER *12 and 14*

### Hard · stretch and reason

7 PROBLEMS

1. A cyclist covers 60 km. Riding 5 km/h faster would save an hour. Find the speed.

ANSWER 15 km/h

2. A train covers 240 km. Going 20 km/h faster would save one hour. Find the speed.

ANSWER  $x^2 + 20x - 4800 = 0$ : 60 km/h

3. A rectangular garden 20 m by 15 m has a path of uniform width around it, doubling the area. Find the width.

ANSWER  $(20 + 2w)(15 + 2w) = 600$ :  $w = 2.5$  m

4. The sum of the squares of two consecutive whole numbers is 313. Find them.

ANSWER  $x^2 + (x + 1)^2 = 313$ : 12 and 13

5. A stone is thrown up and its height is  $h = 20t - 5t^2$ . When is it 15 m high?

ANSWER  $5t^2 - 20t + 15 = 0$ : at  $t = 1$  s going up and  $t = 3$  s coming down.

6. Two workers together finish a job in 6 days. Alone, one takes 5 days longer than the other. Find both times.

ANSWER  $\frac{1}{x} + \frac{1}{x+5} = \frac{1}{6}$ : 10 and 15 days.

7. The area of a triangle is 30 cm<sup>2</sup> and its height is 7 cm more than its base. Find the base.

ANSWER  $\frac{1}{2}x(x + 7) = 30$ :  $x = 5$  cm

## 07 Homework

### Homework — 6 problems

Algebra 8, Chapter III, pp. 153–158. Always say which root you reject and why.

1. A rectangle is 5 cm longer than wide and has area 84 cm<sup>2</sup>. Find its dimensions.
2. Two consecutive whole numbers have product 342. Find them.
3. A right triangle has legs  $x$  and  $x + 2$  and hypotenuse 10. Find the legs.
4. A car covers 180 km. Going 10 km/h faster would save half an hour. Find the speed.

5. *The sum of a number and its square is 56. Find both values.*
6. *A rectangle has perimeter 34 m and area 60 m<sup>2</sup>. Find its sides.*